

PARSHIV KAPOOR

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Education

Indian Institute of Technology, Roorkee

August 2023 – Present

BTech Biosciences and Bioengineering

Silver Line Prestige School, Ghaziabad

Senior Secondary Education 90.60%; Physics, Mathematics and Computer Science.

Silver Line Prestige School, Ghaziabad

Secondary School Certificate 94.00%; Physics, Mathematics and Computer Science.

Experience

AI/ML Intern | WhyMinds Global

June 2025 – July 2025

- Designed and implemented multi-agent systems using CrewAI to autonomously monitor financial regulatory websites
- Extracted key financial incidents and transformed them into structured, human-quality blog content simulating the behavior of a specialized editorial team.
- Tools: CrewAI, Groq API, spaCy, BeautifulSoup, LangGraph

Research and Publications

Parshiv Kapoor, et al "Zero-Shot Vision Language Reasoning via Dual-layer Scene Graph Chain of Thoughts" [AAAI-26 Student Abstract Paper](#)

Parshiv Kapoor, et al "TWIST: Temporal Weakly-Supervised Triplets Recognition in Surgical Videos" [AAAI-26 Student Abstract Paper](#)

Parshiv Kapoor, et al "Weighted Color-Morphology Feature Fusion for Tuberculosis Bacilli Detection from Cytopathology Images" [ICVGIP 2025 Paper](#)

Projects

Deep Learning Model for Detecting Acid-Fast Bacilli in Cytopathology Smears

- Employed an autoencoder's channel-wise reconstruction errors to pick the top three informative channels (L, S, V), then fused them with Sobel edges and CBAM attention in a re-weighted ResNet18 backbone
- Designed an asymmetric focal loss (FP=1.7) combined with dynamic threshold tuning to counteract extreme class skew and prioritize bacilli recall
- Benchmarked against a standard RGB ResNet18 baseline, demonstrating clear gains in detection robustness and localization quality under severe class imbalance

Semi-Supervised Semantic Segmentation with Cross-Consistency Training (CCT) | [GitHub](#)

- Implemented Cross-Consistency Training (CCT) for semi-supervised semantic segmentation by enforcing consistency in encoder outputs through novel perturbations
- Trained a shared encoder and main decoder in a supervised manner while using auxiliary decoders to improve encoder representations through consistency constraints
- Applied pseudo-labeling for weak supervision to enhance segmentation performance

Fine Tuning Deep Learning Models on Astrophysical Datasets | **Physics and Astronomy Club**

- Fine-tuned a pre-trained ResNet-50 model on the GalaxyZoo dataset for galaxy morphology classification
- Evaluated baseline performance using ImageNet weights, then applied supervised fine-tuning and Low-Rank Adaptation (LoRA) for improved accuracy
- Compared performance metrics across different tuning strategies to assess model generalization and domain transfer effectiveness

Generative Adversarial Networks From Scratch | [GitHub](#)

- Implemented foundational Generative Adversarial Networks (GANs) from first principles
- Trained generator and discriminator models in an adversarial framework to synthesize realistic images on the MNIST-fashion dataset
- Developed training loops, loss functions, and evaluation pipelines to explore adversarial learning dynamics and model convergence behavior

Fashion Product Detection and Similarity Search Pipeline **Flickd AI Hackathon** | [GitHub](#)

- Designed a complete fashion search system combining object detection and CLIP to retrieve visually similar products from catalog images
- Automatically generated text prompts and stored image-text embeddings in FAISS to enable fast similarity search
- Built a clear, modular pipeline that connects detection, embedding creation, and retrieval for practical fashion search use cases

Skills

Programming Languages: Python C++

Machine Learning/Deep Learning/NLP: PyTorch, TensorFlow, scikit-learn, Keras, Hugging Face, spaCy, NLTK, LangChain, LlamaIndex, CrewAI, LangGraph

Awards & Achievements

JEE Advanced: All India Rank 8530

JEE Mains: All India Rank 12949

SeePhys Challenge @ ICML 2025: Achieved 5th rank globally in a large-scale vision-language competition on physics reasoning with 2,000+ multimodal problems and 2,200+ diagrams. Built and optimized a VLM-based system capable of solving complex physics questions by combining schematic interpretation with natural language understanding.

Beginner's Hypothesis: Runner Up in competition organized by Data Sciences Group (DSG) at IIT Roorkee

Amazon ML Challenge 2025: Secured a rank of 180 out of around 10,000 participants